LEARNING EQUIVARIANT NON-LOCAL ELECTRON DENSITY FUNCTIONALS

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ABSTRACT

The accuracy of density functional theory hinges on the approximation of non-local contributions to the exchange-correlation (XC) functional. To date, machine-learned and human-designed approximations suffer from insufficient accuracy, limited scalability, or dependence on costly reference data. To address these issues, we introduce Equivariant Graph Exchange Correlation (EG-XC), a novel non-local XC functional based on equivariant graph neural networks (GNNs). Where previous works relied on semi-local functionals or fixed-size descriptors of the density, we compress the electron density into an SO(3)-equivariant nuclei-centered point cloud representation for efficient non-local atomic-range interactions. By applying an equivariant GNN on this point cloud, we capture molecular-range interactions in a scalable and accurate manner. To train EG-XC, we differentiate through a self-consistent field solver requiring only energy targets. In our empirical evaluation, we find EG-XC to accurately reconstruct 'gold-standard' CCSD(T) energies on MD17. On out-of-distribution conformations of 3BPA, EG-XC reduces the relative MAE by 35% to 50%. Remarkably, EG-XC excels in data efficiency and molecular size extrapolation on QM9, matching force fields trained on 5 times more and larger molecules. On identical training sets, EG-XC yields on average 51% lower MAEs.

3 RELATED WORK

Learnable density functional approximations (DFAs) of the XC functional have a long history. In their seminal paper, Kohn & Sham (1965) proposed an ϵ_{XC} fitted to a reference calculation. In this first parameterization ϵ_{XC} only depends on $g(r) = \rho(r)$. To improve accuracy, the density features g have been successively refined by including gradients and physical quantities such as the kinetic energy density $\tau(r) = \frac{1}{2} \sum_{i}^{N_{\mathrm{el}}} |\nabla \phi_i(r)|^2$. The resulting semi-local functionals are also known as meta-GGAs with $g(r) = g_{\mathrm{mGGA}}(r) \coloneqq [\rho(r), |\nabla \rho(r)|, \tau(r)]$ (Perdew, 2001). Many ML functionals use this parameterization with ϵ_{XC} frequently being a product of an MLP with a classical functional (Nagai et al., 2022; Kulik et al., 2022; Zhang et al., 2023). An appealing aspect of learning semi-local energy densities rather than $E_{\mathrm{XC}}[\rho]$ directly is the known implementation of physical constraints (Sun et al., 2015; Kaplan et al., 2023), which inspired works by Dick & Fernandez-Serra (2021); Nagai et al. (2022). While augmenting g_{mGGA} with hand-crafted non-local features improves the accuracy, this comes at the cost of exact constraints (Nagai et al., 2020; Kirkpatrick et al., 2021). Similarly, such physical biases are absent in kernel methods Margraf & Reuter (2021); Bystrom & Kozinsky (2022). Additionally, these methods require reference densities that are rarely available (Szabo & Ostlund, 2012). Alternative grid-based CNNs converge slowly in the number of grid points compared to specialized spherical integration grids (Zhou et al., 2019; Treutler & Ahlrichs, 1995). Lastly, machine-learned functionals have been used in other DFT contexts as well, e.g., for energy corrections on SCF-converged energies (Chen & Yang, 2024) or to model the kinetic energy functional in orbital-free DFT (Snyder et al., 2012; Mi et al., 2023; Zhang et al., 2024). Here, we propose to extend semi-local models with expressive fully learnable non-local features obtained from standard integration grids. This allows us to leverage the physical biases of meta-GGAs while allowing efficient non-local interactions with fast integration. Although this work focuses on the XC functional, the methods we present can be transferred to these applications as well.

Machine learning force fields have long functioned as a cheap but inaccurate alternative to quantum mechanical calculations by directly approximating the potential energy surface from data without solving the electronic structure problem (Unke et al., 2021). Contemporary force fields generally rely on graph neural networks (GNNs) (Schütt et al., 2017) representing molecules in terms of graphs with atoms as nodes. The advent of SO(3)-equivariant models led to significant improvements in accuracy and data efficiency, closing the gap to full-fidelity quantum mechanical calculations (Batzner et al., 2022; Batatia et al., 2022). While they effectively aim to accomplish the same goal as DFT at a fraction of the cost, their out-of-distribution accuracy is a common problem (Stocker et al., 2022). In EG-XC, we transfer the success of equivariant GNNs to DFT, by combining their approximation power with the physical nature of DFT, resulting in an accurate and data-efficient method.

6 DISCUSSION

We tackled the problem of learning efficient non-local exchange-correlation functionals to enhance the accuracy of density functional theory. To this end, we have presented Equivariant Graph Exchange Correlation (EG-XC), an equivariant graph neural network approach. We have introduced a basis-independent reduction of the electron density into a finite point cloud representation, enabling equivariant graph neural networks to capture molecular-range interactions. Based on this point cloud embedding, we have parameterized a non-local feature density used for the reweighing of a semi-local exchange-correlation energy density. Unlike other learnable non-local functionals, EG-XC can be trained by differentiating through the SCF solver such that the training requires only energy targets.

In our empirical evaluation, EG-XC improves upon previous machine learning-based methods of modeling potential energy surfaces. On MD17, EG-XC accurately reconstructs 'gold-standard' potential energy surfaces, namely CCSD(T), within the DFT framework. On the 3BPA, EG-XC reduces errors by 35% to 50% compared to the best-performing baseline. On QM9, EG-XC demonstrates remarkable data efficiency, achieving similar accuracies with 5 times less data or up to 33% lower errors at the same amount of data compared to the best baseline. Finally, EG-XC extrapolates well to unseen conformations and larger molecules on 3BPA and QM9, respectively.

Overall, these results strongly underline the data efficiency of learning exchange-correlation functionals compared to machine-learned force fields. We hypothesize that a significant portion of EG-XC's generalization is thanks to including the physically constrained semi-local model that captures most of the exchange-correlation energy and biases EG-XC to approximately fulfill the same constraints. On top of these, our non-local contributions transfer the accuracies of graph neural network-based force fields to functionals while maintaining a similar data efficiency.

Future work. The simple implementation of EG-XC through standard deep learning libraries and the integration with the self-consistent field method open the door to a range of future research. Thanks to the basis-set independence, EG-XC can transfer to non-atom-centered basis sets like plane waves, as they are common in periodic systems, or approximate the kinetic energy functional in orbital-free DFT. Further, while we trained EG-XC solely on energies, multimodal training with other DFT-computable observables like electron densities or atomic forces could further improve accuracy and generalization. Another path to improving generalization may be integrating known physical constraints to the non-local part of EG-XC. Finally, accurate functionals like EG-XC may bridge the gap between sparse, accurate quantum mechanical calculations (Cheng et al., 2024; Gao & Gunnemann, 2024) and fast force field (Batzner et al., 2022).